IMPLEMENTATION GUIDE

360° AR Capture

How it works, step by step

One station. Twenty-six fixed targets on a 1.5 m sphere around it. Twenty-six photographs, taken one at a time, each aimed at a marker that stays put in the real world while you turn, tilt and step.

What it produces	26 JPEGs plus a station record, one folder per station
Where it runs	Android, ARCore, Flutter UI, native GLES renderer
Targets	26 directions — three rings of eight, plus zenith and nadir
Status of this document	describes the current source; measured figures in section 8

How to read this

Sections 1–3 explain what the app is doing and in what order. Section 4 lists the numbers that decide everything, so you can check any behaviour against them. Sections 5–7 are the reference: what the screen says, what happens when AR loses tracking, and what to do when a step will not complete. Section 8 is the evidence that the arrangement really is fixed in the world.

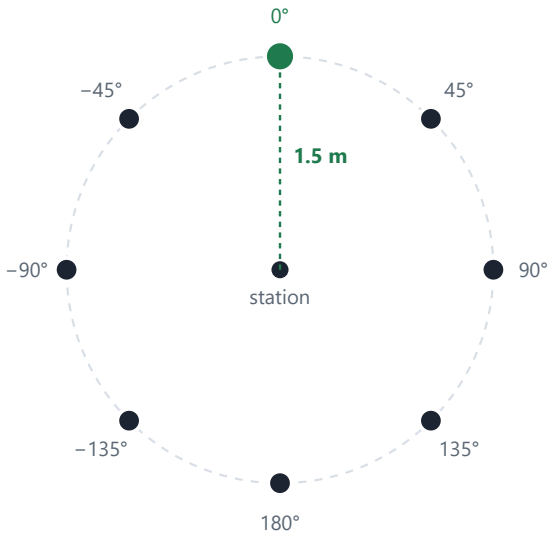
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01 What the app does

You stand somewhere in a room, hold the phone up, and press **Calibrate station**. From that moment the app knows two things: **where** you are standing, and **which way** you were facing. Everything else is derived from those two facts.

It then builds an imaginary sphere of radius 1.5 m around that spot and puts 26 markers on it — eight around the horizon, eight 45° above, eight 45° below, one straight up and one straight down. Each marker is pinned to a real position in the room with an ARCore world anchor, and stays there for the rest of the session. Your job is to point the phone at each one in turn and hold still; the app takes the photograph, saves it, and moves you to the next.



T01	front — first in the sequence	**T17–T24**	lower ring, −45°
T02–T08	rest of the middle ring, 0°	**T25**	zenith, +90°
T09–T16	upper ring, +45°	**T26**	nadir, −90°

The ring drawn is the middle one; the other two sit 45° above and below it.
At calibration you face the 0° direction, so the front target comes first.

Figure 1 — the arrangement seen from above. The sequence starts at the front target, because that is where you were already looking when you pressed Calibrate.

Twenty-six is not arbitrary: three rings of eight plus two poles covers the whole sphere with enough overlap that a stitched panorama has no gaps, and it is small enough to finish in one sitting.

02 The words this document uses

Term	What it means here
Station	The one spot you calibrated from. Its position and orientation are frozen and never re-derived.
Target	One of the 26 directions, and the marker placed 1.5 m along it.
Anchor	ARCore's handle on a real-world position. The marker is drawn wherever the anchor is.
Plate / bezel	The two GLB models: a white disc with a green centre, and the ring around it.
Ring	The reticle in the middle of the screen — the same shape as the bezel. It fills as you hold still.
Alignment	The 3-D angle between where the camera points and where the target is.
Stability window	How long the phone must stay aimed and still before the shutter may fire.
Drift	How far you have walked from the station. Past 0.6 m the shutter is blocked.
Ready	Every condition is met. Only then can a photograph be taken.

03 The working sequence

Nine steps, in the order the app performs them. Steps 5 to 7 repeat 26 times — once per target.

1. **Boot the AR session**
   ARCore starts; the camera frame reaches the screen
2. **Calibrate the station**
   freeze a level reference at your position
3. **Place all 26 targets**
   one world anchor each, 1.5 m out, never moved again
4. **Aim at the current target**
   arrow + sentence point at the marker
5. **Hold still**
   the ring fills while the pose stays steady
6. **Ring completes**
   target hides, shutter fires by itself
7. **Save and verify**
   write, read back, record — or fail and re-arm
8. **Advance**
   the next pending target becomes active and visible
9. **All 26 done**
   the station is complete

Step 1 — Boot the AR session

What happens. The app asks ARCore for a camera session, creates the texture the preview renders into, and starts listening to the pose stream — about 30 to 60 position-and-orientation readings a second.

What you see. The preview appears. The status line reads **AR starting**, then **AR tracking OK** once ARCore has found the room.

Why it works that way. Nothing is placed yet. Placement needs a station, and a station needs a pose the app can trust.

Step 2 — Calibrate the station

What happens. You press **Calibrate station**. The app takes the current camera pose and rebuilds it from gravity: **up** comes from the world's vertical axis, **forward** is the direction you are facing flattened onto the horizontal plane. Your pitch and roll are deliberately thrown away.

What you see. The status line reports how many targets were placed. The first target's marker appears straight ahead.

Why it works that way. If the reference kept the phone's tilt, the whole sphere would lean by whatever angle you happened to hold it at. Levelling it against gravity means the rings sit level however you were holding the phone. If you are pointing within 10° of straight up or down, calibration is refused — there is no reliable forward direction there.

Step 3 — Place all 26 targets

What happens. For each of the 26 directions the app computes one world position — station, plus 1.5 m along that direction — and asks ARCore to hold a world anchor there. All 26 are created in this single pass and kept for the whole session.

What you see. The active target is solid; the other 25 are faint ghosts at their own positions.

Why it works that way. The position is **arithmetic, not a raycast**. No wall, floor or ceiling is consulted, so an unmapped room, an empty unit or a bare wall all place the targets identically. That is also why the shutter never waits for surfaces to be found.

Step 4 — Aim at the current target

What happens. Every frame the app compares the camera's forward axis with the target's current position and produces two things: a sentence, and an arrow. The arrow is drawn towards the target's actual place on the screen, not towards a bearing.

What you see. Aim at the front target. Turn right 45°, with a short arrow pointing that way. When the target is off screen the arrow becomes an edge marker on the correct side.

Why it works that way. The arrow uses the same projection as the renderer, so it points at the marker you can see. It disappears once you are inside tolerance — at that point the ring is the whole message.

Step 5 — Hold still

What happens. The ring starts filling. It only fills while the alignment is inside 3° and the phone is genuinely still: no more than 5°/s of rotation, no more than 3 cm of wandering, readings no older than 250 ms.

What you see. The ring sweeps clockwise in green. The chip reads **Hold still**, then **Ready**.

Why it works that way. One small tremor should not restart the wait, so the window tolerates 2° of wobble beyond the 3° tolerance before it gives up. A real turn breaks it immediately.

Step 6 — The ring completes

What happens. The ring is full. The app hides the active marker and its bezel — they are simply left out of the drawing — and fires the shutter. The anchor is untouched.

What you see. The marker vanishes and the chip reads **Validating image**.

Why it works that way. Hiding it is the point: a full ring means the photograph is being taken, so a marker still sitting in front of the lens is only in the way. If the photograph fails, the marker comes back and you re-aim.

Step 7 — Save and verify

What happens. Five stages, each with its own deadline: take the still (6 s), write the file (10 s), check it exists, read it back and compare its length, then write the record (10 s). Any failure stops the sequence and says which stage it stopped at.

What you see. A confirmation line with the image size, then the next target becomes active.

Why it works that way. The target is only marked captured after the file is on disk **and** read back intact. A write that returned but produced a truncated file is not a photograph.

Step 8 — Advance

What happens. The app steps to the next target that has no photograph, starting from the one just accepted. If you captured them out of order, it follows the list, not the numbering.

What you see. The next marker becomes solid at its own position. Every completed marker stays hidden.

Why it works that way. Only the highlight moves. No target is created, moved or released, so the arrangement you calibrated against is the arrangement you finish with.

Step 9 — All 26 done

What happens. When the 26th photograph is accepted, the station is complete.

What you see. Every marker is gone; the screen offers the review.

Why it works that way. The station's folder holds the reference pose, the 26 targets, the 26 records and the 26 JPEGs.

What one target costs you

Phase	Typical time	What sets it
Aim and turn	2–5 s	how far you have to rotate
Hold still	0.45 s	the stability window, fixed
Shutter and save	0.1–0.3 s	camera handover, file write, read-back
Advance	instant	the next marker becomes active

Twenty-six targets at a few seconds each is a two-to-three minute session, plus whatever time ARCore needs to find the room at the start. The bottleneck is almost always the aim, not the app.

04 The numbers that decide everything

Every behaviour above comes from a handful of constants. If something feels wrong, this is the table to check it against — they are the defaults in `CaptureGeometryConfig` and `PhotoCaptureService`.

Setting	Value	What it controls
Target radius	1.5 m	How far each target sits from the station.
Plate diameter	0.22 m	The physical size of the white disc — about 8° of view at 1.5 m.
Draw scale	0.407×	0.22 ÷ 0.54 (the plate's authored width). Applied once, to plate and bezel alike.
Bezel setback	6 mm	How far the ring sits behind the plate so the two faces are flush.
Alignment tolerance	3.0°	The shutter's strict test, and the angle at which the arrow disappears.
Window hysteresis	+2.0°	How far past tolerance you may wobble before the hold is thrown away.
Stability window	450 ms	How long you must stay aimed and still.
Max angular rate	5 °/s	Faster than this and the hold does not start.
Max orientation drift	1.5°	Total turning allowed across the window.
Max position drift	3 cm	Total wandering allowed across the window.
Max reading age	250 ms	An older pose is not acted on.
Station drift limit	0.6 m	Past this the shutter is blocked and the app asks you to walk back.
Still timeout	6 s	How long the camera may take to hand over an image.
Save timeout	10 s	Per write and per record.
Minimum JPEG	2,048 bytes	Anything smaller is not treated as a photograph.

Two of these are easy to confuse

3° is the shutter's test at the instant of capture, and it is strict. **5°** (3 + 2) is where the stability window gives up. That gap is why a hand tremor does not send you back to the start, while a deliberate turn does.

05 The coordinate rules

Everything in the app is expressed in one frame, anchored at the station: **+Z is forward**, **+X is to your right**, **+Y is up**. Yaw turns around the vertical axis, pitch tilts up and down.

Yaw / pitch	Means	Lands at
0° / 0°	straight ahead	T01 — the front target, first in the sequence
+90° / 0°	90° to your right	T03, on the horizon to your right
180° / 0°	directly behind you	T05
−90° / 0°	90° to your left	T07
any / +45°	halfway up	the upper ring, T09–T16
any / −45°	halfway down	the lower ring, T17–T24
— / +90°	straight up	T25, the zenith
— / −90°	straight down	T26, the nadir

At the two poles the yaw term vanishes: every yaw gives the same point, which is why the guidance at the zenith and nadir uses the plain 3-D angle and never says "turn right".

One detail that looks like a bug and is not: a camera's own right-hand direction is the mirror of a mathematical rotation's, so the app negates the horizontal axis when it converts. The visible consequence is that a target at yaw +90° ("to your right") sits on the world's −X axis. That is correct, and it is what makes the arrow point the way the sentence says.

06 What the screen is telling you

Four places carry information. When something is not working, read them in this order: the status line first, then the phase chip, then the sentence, then the drawer.

Status line — top left

Reads	Means
AR tracking OK · no planes yet	Tracking is fine. No surfaces have been found — irrelevant, because targets are placed by calculation.
AR starting	The session is up but no pose has arrived yet.
AR tracking paused	ARCore has lost confidence. It is still running and will usually recover; nothing is destroyed.
AR tracking stopped	The session has to be restarted. The Retry action appears.

Phase chip — the small label near the shutter

Chip	Means	Can you shoot?
Initialising	No trustworthy pose yet.	No
Aim at target	A pose, but not aimed — or aimed from the wrong place.	No
Hold still	Aimed, ring filling.	No
Ready	Aimed, steady, every condition met.	Yes — and it fires by itself
Validating image	A photograph is being written.	No
Accepted	Saved and recorded.	—
Retry	The last attempt failed. Re-aim to try again.	No

The sentence in the middle

This is the instruction, and it names the action: **Aim at the front target. Turn right 45°**, **Walk back to the station, or press Recalibrate here**, **The last attempt failed — re-aim at the target to try again**. It is derived from the same state as the chip, so the two cannot disagree.

The drawer — tap the status line

Target id and its yaw/pitch, whether the marker is placed, the plane breakdown, the GLB triangle count, the camera's still resolution and focus mode, the plate's physical size and its current distance, and the actionable advice. That last line is the one to read when tracking misbehaves.

07 When AR loses tracking

ARCore reports four states. Only one of them is a problem, and even that one is usually temporary.

State	What the app does
tracking	Normal operation. Targets are drawn from their anchors; the shutter is live once aimed.
paused	The markers stay exactly where they are and keep being drawn — a pause is ARCore losing confidence in the room, not the marker moving. No alignment is computed from the paused pose, the drift figure holds its last trustworthy value, and the shutter is blocked.
stopped	The session must be restarted. The Retry action appears; the station is discarded because a new session has a new world origin.
unavailable	The device cannot run AR. The app says so instead of pretending.

ARCore also says **why** it paused, and the app repeats it verbatim because each reason is a different action:

ARCore's reason	What to do
INSUFFICIENT_FEATURES	Point at something with visible detail — a textured wall, furniture, a door frame. A blank wall gives the tracker nothing to hold on to.
INSUFFICIENT_LIGHT	Move somewhere brighter.

ARCore's reason	What to do
EXCESSIVE_MOTION	Slow down; hold the phone still for a moment.
BAD_STATE	Point back roughly where you started and let it recover.
CAMERA_UNAVAILABLE	Another app is using the camera.

The one thing the app cannot fix

When ARCore has too few features to track, its map of the room is unreliable — and the markers move with that map, because that is where they are anchored. The app can keep them placed, keep them drawn, refuse to take a photograph from an unreliable world, and tell you why. It cannot make the tracker see detail that is not there. More texture and more light are the only real remedies.

08 Evidence that the arrangement really is fixed

The claim "the targets do not move with the phone" is testable without a device, because the position of every target is computed from the station alone. `dart run tool/verify_sphere.dart` places all 26 through the same function the app uses, then moves the camera four different ways and measures how far the targets moved.

Check	Result
Distance from the station, all 26 targets	1.500 m — worst error 2.2×10^{-16} m (floating-point noise)
Yaw recovered from the placed position	matches the spec exactly — worst error 0.0°
Pitch recovered from the placed position	worst error 7.1×10^{-15}°
World movement, camera at the station / turned 90° / tilted 40° / walked 2 m	0.0 m — literally zero
Marker facing as the camera walks away	fixed (facing·toCamera falls 1.00 → 0.24) — not a billboard

What does change is where the target appears on screen, and that is the perspective you are meant to see:

Camera	T01 on a 1080×2400 screen
at the station, facing forward	dead centre (540, 1200)
turned 90° right	behind you — pinned to the edge (1064, 1200)
tilted up 40°	off the bottom (540, 2693)
walked 2 m right and 1 m forward	off to the left (−407, 822)

The same run is also the quickest way to confirm, after any change, that the geometry has not regressed: the last line must read `0.0e+0 m`.

09 When a step will not complete

Symptom	Likely cause	What to do
"AR tracking paused" and it stays that way	Too few features or too little light for the tracker	Point at textured surfaces; add light. The markers are still placed — nothing has been lost.

Symptom	Likely cause	What to do
Ring never fills	The phone is moving, or the reading is stale	Hold still; the sentence names the reason ("moving at 8.2°/s", "waiting for a fresh pose").
Ring fills, chip says Ready, nothing happens	A capture is already in flight	Wait for the confirmation line. Only one photograph is taken at a time.
"Walk back to the station"	You are more than 0.6 m from where you calibrated	Walk back, or press Recalibrate here to start a new station where you stand.
"The last attempt failed — re-aim"	The camera or the disk refused the photograph	Look away and back, then try again. The error line says which stage failed.
A marker vanished mid-sequence	Its photograph was accepted, or you are aimed and steady (the ring is full)	Expected. It comes back if the hold is lost or the photograph fails.
Targets look bigger or smaller than each other	Perspective — they are at different distances from the lens	Nothing to fix. Every target is 0.22 m across; the drawer shows the size and the distance.
Marker sits slightly off the wall it should be on	ARCore's map is still refining, or it is drifting for lack of features	Move the phone to map more of the room. The marker follows the anchor, not the wall.

10 Where the code lives

File	Responsibility
capture_controller.dart	The conductor: calibration, placement, the pose loop, the shutter, advancing.
capture_state.dart	The one immutable snapshot the screen paints, and the single readiness decision.
target_sphere.dart	The 26 directions, the geometry constants, and resolveTarget — the one authority for placement.
station_frame.dart	The frozen station reference and the gravity-aligned calibration.
plane_anchor_service.dart	The 26 world anchors, the scene push, and the hidden/visible set.
alignment_calculator.dart	Camera pose → target position → the angle, the turn, the tilt.
stability_monitor.dart	The 450 ms window and its five conditions.
photo_capture_service.dart	Take, write, read back, record — with a deadline per stage.
projection.dart	The pinhole model the arrow and the marker both use.
ArCoreSessionController.kt	The native side: session, render loop, anchors, stills.
ArGlRenderer.kt	The GLES renderer: mvp = projection × view × anchor × rotation × offset × scale.
tool/verify_sphere.dart	The measurement in section 8.

This document describes the source as it stands. It is generated from the same constants the app runs on, and the measurements in section 8 are reproduced by running the tool named there — not asserted.